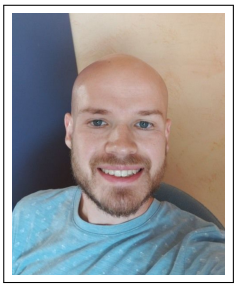
Thomas Jacumin

Born on 5th July, 1995
French citizenship
41 Avenue Georges Clémenceau
06000 Nice
France
+33 6 61 69 78 29
thomas.jacumin@inria.fr



Experiences

September 2025–August 2027 **Postdoc in Applied Mathematics**, *University Côte d'Azur*, France, under the supervision of Pr. R. Masson

March 2023–March 2025 **Postdoc in Applied Mathematics**, *University of Lund*, Sweden, under the supervision of Dr. A. Langer

October 2019–December 2022 **PhD in Applied Mathematics**, *University of Haute–Alsace*, France, under the supervision of Pr. Z. Belhachmi

Education

2019–2022 **PhD in Applied Mathematics**, *University of Haute–Alsace*, France, under the supervision of Pr. Z. Belhachmi

2017–2018 **Master in Modeling and Applied Mathematics**, *University of Augsburg*, Germany, with high honours

2016–2018 **Master in Applied Mathematics**, *University of Rouen*, France, with honours

2015–2016 **Bachelor of Science in Engineering Science**, *École nationale Supérieure en Informatique, Systèmes Avancés et Réseaux*, Valence, France

2013–2015 **DEUG of Mathematics "Prépa Écoles Ingénieurs"**, *University of Avignon*, France, with honours

2010–2013 **Baccalauréat Scientifique**, *Lycée de l'Arc*, Orange, France, with high honours

Publications

- Zakaria Belhachmi and Thomas Jacumin. *Optimal Transport Model of Optical Flow Estimation: Constant and Varying Illumination Cases (accepted in Journal of Mathematical Study)*. Dec. 2024.
- Thomas Jacumin and Andreas Langer. "An Adaptive Finite Difference Method for Total Variation Minimization". In: *Numerical Algorithms* (Mar. 19, 2025). ISSN: 1017-1398, 1572-9265. DOI: 10.1007/s11075-025-02044-6.
- Zakaria Belhachmi and Thomas Jacumin. "Adjoint Method in PDE-Based Image Compression". In: *Asymptotic Analysis* (Oct. 2024), pp. 1–28. ISSN: 18758576, 09217134. DOI: 10.3233/ASY-241944.
- Zakaria Belhachmi and Thomas Jacumin. "Optimal Interpolation Data for PDE-Based Compression of Images with Noise". en. In: *Communications in Nonlinear Science and Numerical Simulation* 109 (June 2022), p. 106278. ISSN: 1007-5704. DOI: 10.1016/j.cnsns.2022.106278.

Proceedings

- Thomas Jacumin et al. "A Coats-Based Formulation for Multiphase Reactive Transport: Application to Hydrogen Storage in Porous Media". In: *ECMOR 2026*. ECMOR 2026. Porto, Portugal, European Association of Geoscientists & Engineers, 2026, pp. 1–20. DOI: 10.3997/2214-4609.202637078.

Preprints

- Zakaria Belhachmi and Thomas Jacumin. *Iterative Approach to Image Compression with Noise: Optimizing Spatial and Tonal Data*. Sept. 29, 2022. DOI: 10.48550/arXiv.2209.14706. arXiv: 2209.14706.

Communications

October 2026 **A Coats-Based Formulation for Multiphase Reactive Transport: Application to Hydrogen Storage in Porous Media**, *RT Terre et Énergies*, Lacanau, France
https://rtte2026cnrs.sciencesconf.org/

September 2026 **A Coats-Based Formulation for Multiphase Reactive Transport: Application to Hydrogen Storage in Porous Media**, *ECMOR 2026*, Porto, Portugal
https://eage.eventsair.com/ecmor-2026/

July 2026 **A Coats-Based Formulation for Multiphase Reactive Transport: Application to Hydrogen Storage in Porous Media**, *CMWR 2026*, Bologna, Italy
https://cmwrconference.org/

March 2026 **Adjoint Method in PDE-Based Image Compression**, *SMAI-MODE 2026*, Nice, France
https://mode2026.sciencesconf.org/

March 2026 **Poster – A Coats-Based Formulation for Multiphase Reactive Transport: Application to Hydrogen Storage in Porous Media**, *APPRISE Winter School*, Aussois, France
https://apprise.sciencesconf.org

February 2026 **PDE-Based Images Compression**, *PhD Seminar*, University Côte d'Azur, France

November 2024 **An Adaptive Finite Difference Method for Total Variation Minimization**, *Seminar*, University of Lund, Sweden

January 2024 **Modèle Transport Optimal pour la Formulation du Flot Optique**, *ANR SOCOT*, University of Haute-Alsace, France
https://codimd.math.cnrs.fr/s/vKWD_JtAu

July 2023 **Méthodes Adaptatives pour les Problèmes de Discontinuité**, *MAThEOR Days*, University of Strasbourg, France
https://days.matheor.com/2023/

October 2022 **Mini-symposium – Inverse problems and shape optimization**, *Optimal interpolation data for PDE-based compression of images with noise*, PICOF22 – University of Caen, France
https://picof22.sciencesconf.org/

June 2022 **PhD Day – "Ma thèse en 180 secondes" format**, University of Haute–Alsace, France

June 2022 **PhD Seminar**, *IRIMAS*, University of Haute–Alsace, France

December 2021 **Poster**, *Calculus of Variations Colloquium*, University of Nancy, France
https://indico.math.cnrs.fr/event/7000/

Teaching

September–December 2026 **Méthodes d'Algèbre Linéaire 1**, *Exercice sessions intended for first-year undergraduate students in Applied Mathematics and Computer Science for the Social Sciences*, University of Côte d'Azur, France

January–April 2026 **Calculus 2**, *Exercice sessions intended for first-year undergraduate students in Applied Mathematics and Computer Science for the Social Sciences and Mathematics*, University of Côte d'Azur, France

March–November 2024 **Bachelor Thesis Supervision**, *Adaptive Finite Differences Method*, University of Lund, Sweden

April–May 2022 **Master 1 Project Supervision**, *Image Compression by Image Inpainting*, University of Haute-Alsace, France

Work Groups

November 2022 **Benamou-Brenier Formula**, *Optimal Transport*, University of Haute–Alsace, France
https://juillet.perso.math.cnrs.fr/GTTO.html

March 2022 **C–transforms**, *Optimal Transport*, University of Haute–Alsace, France

Attended Schools

March 2026 **Winter school APPRISE (Fundamentals of trAnsport Phenomena in PoRous medIa across ScalEs)**, Centre Paul-Langevin, France

July 2023 **Summer school on Numerical Analysis of nonlinear PDEs**, University of Leipzig, Germany

Other Experiences

August 2022 **PhD Seminar Organization**, *Website Development*, IRIMAS, University of Haute–Alsace

June–July 2022 **Colloquium Organization**, *Website Development*, IRIMAS, University of Haute–Alsace
`http://www.lmia.uha.fr/colloque-lutz/`

June 2022 **Event Organization**, *Music Festival*, University of Haute–Alsace

September 2021 **AMIES Challenge**, *Trajectories Reconstruction*, Eurecam

Skills

French Native

English Good

Japanese Notions

IT Tools Python, SageMath, MATLAB, FreeFem++, Latex, C, C++, Java, VHDL, CUDA, Fortran, OpenFOAM

Referees

Belhachmi Z.,	Professor,	IRIMAS,	*University of Haute–Alsace*,	`zakaria.belhachmi@uha.fr`
Langer A.,	Assistant Professor,	LTH,	*University of Lund*,	`andreas.langer@math.lth.se`
Masson R.,	Professor,	LJAD,	*University Côte d'Azur*,	`roland.masson@univ-cotedazur.fr`